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Center side airbag

Abstract

The present invention relates to a center side airbag capable of supporting the airbag cushion without relying on the center console to protect the occupants. The present invention introduces a center side airbag including an airbag cushion arranged between a first seat and a second seat and configured to deploy upward and downward, a lower region configured to deploy in a shape protruding toward the second seat from a lower portion of the airbag cushion, and an upper region configured to deploy in a shape protruding toward an occupant seated on the first seat from an upper portion of the airbag cushion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application claims priority to Korean Patent Application No. 10-2022-0139464, filed Oct. 26, 2022 and Korean Patent Application No. 10-2022-0140612, filed Oct. 27, 2022, the entire contents of which are incorporated herein for all purposes by this reference.

TECHNICAL FIELD

(2) The present invention relates to a center side airbag that supports the airbag cushion without relying on the center console to protect the occupants.

BACKGROUND

(3) The center side airbag deploys vertically between the driver and the front passenger, and it is

supported either by the center console or between the driver and the front passenger to prevent collisions between occupants and to mitigate secondary collisions.

(4) However, in electric vehicles or situations where the center console is removed or movable, it is not feasible to rely on the center console to support the airbag cushion.

(5) In this case, when the occupants load onto the cushion, there is a problem with the cushion rotating horizontally and/or vertically around the mounting point on the seat, which significantly diminishes the restraint force applied to the occupants.

(6) While tethers can assist in shaping the deployment of the cushion, relying solely on tethers poses challenges in effectively restraining the occupants.

(7) The foregoing is intended merely to aid in the understanding of the background of the present invention, and is not intended to mean that the present invention falls within the purview of the related art that is already known to those skilled in the art.

SUMMARY

(8) The present invention has been conceived to solve the above problems and it is an object of the present invention to provide a center side airbag capable of supporting the airbag cushion without depending on the center console for protecting the occupants.

(9) In order to accomplish the above objects, the present invention is characterized by including an airbag cushion arranged between a first seat and a second seat and configured to deploy upward and downward, a lower region configured to deploy in a shape protruding toward the second seat from a lower portion of the airbag cushion, and an upper region configured to deploy in a shape protruding toward an occupant seated on the first seat from an upper portion of the airbag cushion. the lower region may deploy in a wrapping motion protruding in one or more of inward and outward directions with respect to a side of the first seat.

(10) The lower region may be wrapped by the tension created by a connecting member connected between an end portion of the lower region and an intermediate region of the airbag cushion.

(11) The upper region may deploy in a wrapping motion protruding in one or more of inward and outward directions with respect to a side of the first seat.

(12) The upper region may be wrapped by the tension created by a connecting member connected between an end portion of the upper region and an intermediate region of the airbag cushion.

(13) The intermediate region may include an intermediate inactive zone without being filled with gas, and the connecting member is connected to the intermediate inactive zone.

(14) The connecting member may be a tether. the connecting member may be an upper and lower inactive zone formed to extend from an end portion of the upper region and an end portion of the lower region.

(15) The lower region of the airbag cushion may deploy prior to the upper region.

(16) The airbag cushion may include a first cushion deploying from the lower region and a second cushion deploying from the upper region, the first cushion deploying between a side of a first seat and a side of a second seat, the second cushion being connected to the first cushion to deploy upward at a position supporting a head of an occupant seated on the first seat, a connecting member being connected to the second cushion to create tension to pulling the second cushion toward the occupant seated on the first seat.

(17) The first cushion may be supported by a seat cushion of the first seat.

(18) The first cushion may deploy from the rear end of a side of a seat cushion.

(19) The second cushion may be connected to a position biased toward the first seat on the first cushion.

(20) The first cushion may deploy prior to the second cushion.

(21) The second cushion may be smaller in deployment capacity than the first cushion.

(22) The first cushion and the second cushion may be connected for gas to flow therebetween.

(23) The connecting member may be connected to a position supporting the occupant on the first seat in response to the airbag being operated.

- (24) The connecting member may be connected between the first cushion and the second cushion.
- (25) The connecting member may be connected between a side of the first cushion facing the first seat and the upper end of the second cushion.
- (26) The connecting member may be a tether.
- (27) Through the means of solving the problems, the present invention achieves the advantageous effect of protecting the occupants' safety by providing support to the lower region of the airbag cushion on the front passenger seat in vehicles without a center console, which restrains the driver's legs and allows the upper region of the airbag cushion simultaneously restrains the driver's head.
- (28) Furthermore, supporting the airbag cushion on the front passenger seat prevents the rotation of the airbag cushion, which prevent the upper body of the driver from shifting towards the front passenger side, resulting in an improvement in the restraint force on the occupants.
- (29) Furthermore, by supporting the occupants solely through the deformation of the airbag cushion's shape, there is no need to increase the volume of the cushion or the capacity of the inflator, resulting in cost and weight reduction of the airbag cushion and offering favorable advantages in terms of airbag packaging.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram illustrating the deployed state of a center side airbag on a seat according to the present invention.
- (2) FIG. 2 is a diagram illustrating the operational state where a center side airbag is supported on the front passenger seat while supporting the driver's seat occupant;
- (3) FIGS. 3 to 5 are diagrams illustrating a lower region of an airbag cushion protruding in different shapes according to different embodiments of the present invention;
- (4) FIGS. 6 to 8 are diagrams illustrating an upper region of an airbag cushion protruding in different shapes according to different embodiments of the present invention;
- (5) FIG. 9 is a diagram illustrating a deployed airbag cushion with tethers connected thereto according to the present invention;
- (6) FIG. 10 is a diagram illustrating a deployed airbag cushion with upper and lower inactive zones according to the present invention;
- (7) FIG. 11 is a diagram illustrating the operational state of an airbag supported on the inner side of the vehicle cabin is supporting the driver's seat occupant according to the present invention;
- (8) FIG. 12 is a diagram illustrating the operational state of the center side airbag supported on the front passenger seat is deployed along with first and second cushions responsible for upper and lower regions to support the driver's seat occupant according to the present invention;
- (9) FIG. 13 is a diagram illustrating the deployment positions of the first cushion and the second cushion according to the present invention; and
- (10) FIG. 14 is a diagram illustrating the operational state of an airbag supported on the inner side of the vehicle cabin is supporting the driver's seat occupant according to the present invention.

DETAILED DESCRIPTION

- (11) Hereinafter, descriptions are made of the embodiments disclosed in the present specification with reference to the accompanying drawings in which the same reference numbers are assigned to refer to the same or like components and redundant description thereof is omitted.
- (12) As used in the following description, the suffixes “module” and “unit” are granted or used interchangeably for convenience of description but do not have distinct meanings or roles by themselves.
- (13) In addition, detailed descriptions of well-known technologies related to the embodiments disclosed in the present specification may be omitted to avoid obscuring the subject matter of the

embodiments disclosed in the present specification. In addition, the accompanying drawings are only for easy understanding of the embodiments disclosed in the present specification and do not limit the technical spirit disclosed herein, and it should be understood that the embodiments include all changes, equivalents, and substitutes within the spirit and scope of the disclosure.

(14) As used herein, terms including an ordinal number such as “first” and “second” can be used to describe various components without limiting the components. The terms are used only for distinguishing one component from another component.

(15) It will be understood that when a component is referred to as being “connected to” or “coupled to” another component, it can be directly connected or coupled to the other component or intervening component may be present. In contrast, when a component is referred to as being “directly connected to” or “directly coupled to” another component, there are no intervening components present.

(16) As used herein, the singular forms are intended to include the plural forms as well, unless the context clearly indicates otherwise.

(17) It will be further understood that the terms “comprises” or “has,” when used in this specification, specify the presence of a stated feature, number, step, operation, component, element, or a combination thereof, but they do not preclude the presence or addition of one or more other features, numbers, steps, operations, components, elements, or combinations thereof.

(18) With reference to FIGS. 1 and 2, the center side airbag of the present invention includes an airbag cushion **100** that deploys vertically between a first seat **10** and a second seat **20** arranged side by side. The airbag cushion **100** includes or is divided into (1) a lower region (or lower cushion) **110** that protrudes towards the second seat **20** from the lower part of the airbag cushion **100** when deployed, (2) an upper region (or upper cushion) **120** that protrudes towards a passenger seated on the first seat **10** from the upper part of the airbag cushion **100** when deployed, and (3) an intermediate region **130** extending vertically between the lower region **110** and the upper region **120**. The airbag cushion **100** is configured such that, when inflated and deployed between the first seat **10** and the second seat **20**, the intermediate region **130** has a first length in a vertical direction of a vehicle that is greater than a second length of the lower region **110** in the vertical direction of the vehicle and a third length of the upper region **120** in the vertical direction of the vehicle.

(19) For example, the first seat **10** may be a driver's seat, while the second seat **20** may be a front passenger seat.

(20) The airbag cushion **100** of the center side airbag is fixed to the side of the seat back frame of the driver's seat, deploying between the driver's seat and the front passenger seat.

(21) The airbag cushion **100** is formed in a narrow width and long length chamber shape and deploys in the vertical direction.

(22) The lower region **110** of the airbag cushion **100**, which deploys from the lower side position of the seat, is deployed in a shape protruding towards the front passenger seat to be supported on the side of the front passenger seat.

(23) The upper region **120** of the airbag cushion **100**, which deploys from the upper side position of the seat, is deployed protruding towards the driver seated on the driver's seat, providing support to the upper body of the driver.

(24) Therefore, in vehicles without a center console, the lower region **110** of the airbag cushion **100** is supported on the front passenger seat, restraining the lower body of the driver, while the upper region **120** of the airbag cushion **100** restrains the head of the driver, providing protection to the driver.

(25) Furthermore, the airbag cushion **100** supported on the front passenger seat prevents the rotation of the airbag cushion **100** when a passenger is loaded onto the airbag cushion **100**.

(26) Therefore, it prevents the upper body of the driver from shifting towards the front passenger side, thus enhancing the restraint force on the driver.

(27) In the present invention, the lower region **110** may deploy in a wrapping motion protruding in

one or more of inward and outward directions with respect to a side of the first seat **10**. For example, the lower region **100** may be bent when deployed and protrude toward the second seat **20**.

(28) As a preferable example, as shown in FIG. 3, the end portion of the lower region **110** of the airbag cushion **100** may deploy in a wrapping motion in an outward direction towards the front passenger seat.

(29) As another example, as shown in FIG. 4, the end portion of the lower region **110** of the airbag cushion **100** may deploy in a wrapping motion in an inward direction towards the driver's seat.

(30) As another example, as shown in FIG. 5, the end portion of the lower region **110** of the airbag cushion **100** may deploy in a wrapping motion in an outward direction towards the front passenger seat and simultaneously in an inward direction towards the driver's seat.

(31) The above embodiments differ in the direction in which the lower region **110** of the airbag cushion **100** wraps but identical in that the lower region **110** protrudes towards the front passenger seat and is supported by the front passenger seat.

(32) As shown in FIG. 1, a first connecting member **200** may be connected between the end portion of the lower region **110** and the intermediate region **130** of the airbag cushion **100**, creating tension to allow the lower region **110** to wrap.

(33) For example, one end of the first connecting member **200** may be fixed to the end portion of the lower region **110** of the airbag cushion **100**, and the other end of the first connecting member **200** may be pulled upward and fixed to the outer middle portion of the airbag cushion **100**, allowing the lower region **110** to deploy in a wrapped shape outward.

(34) Meanwhile, in the present invention, the upper region **120** may deploy in a wrapping motion protruding in one or more of inward and outward directions with respect to a side of the first seat **10**. For example, the upper region **120** may be bent when deployed and protrude toward the first seat **10**.

(35) As a preferred example, as shown in FIG. 6, the end portion of the upper region **120** of the airbag cushion **100** may deploy in a wrapping motion in an inward direction towards the driver's seat.

(36) As another example, as shown in FIG. 7, the end portion of the upper region **120** of the airbag cushion **100** may deploy in a wrapping motion in an outward direction towards the front passenger seat.

(37) As another example, as shown in FIG. 8, the end portion of the upper region **120** of the airbag cushion **100** may deploy in a wrapping motion in an inward direction towards the driver's seat and simultaneously in an outward direction towards the front passenger seat.

(38) The above embodiments differ in the direction in which the upper region **120** of the airbag cushion **100** wraps, but identical in that the upper region **120** protrudes towards the driver's seat, providing support to the driver.

(39) As shown in FIG. 1, a first connecting member **200** may be connected between the end of the upper region **120** and the intermediate region **130** of the airbag cushion **100**, creating a tension and allowing the upper region **120** to wrap.

(40) For example, one end of the first connecting member **200** may be fixed to the end portion of the upper region **120** of the airbag cushion **100**, and the other end of the first connecting member **200** may be pulled downward and fixed to the inner middle portion of the airbag cushion **100**, allowing the upper region **120** to deploy in a wrapped shape inward.

(41) Furthermore, as shown in FIGS. 1 and 9, an intermediate inactive zone **140**, which is not filled with gas, may be formed in the intermediate region **130**, and the first connecting member **200** may be connected to the intermediate inactive zone **140**.

(42) For example, the intermediate inactive zone **140** is formed along the vertical length direction of the airbag cushion **100**, and by securing the first connecting member **200** within the intermediate inactive zone **140**, the first connecting member **200** may be securely fixed to the airbag cushion **100**.

(43) In this case, the amount of bending and protrusion of the lower region **110** and upper region **120** may be adjusted based on the length of the intermediate inactive zone **140** or the length of the first connecting member **200**.

(44) As an example of the first connecting member **200**, a tether **210** may be employed.

(45) The tether **210** is formed in the shape of a belt or band of which one end is sewn to the end portion of the lower and upper regions **110** and **120** and the other end of each tether **210** is sewn to the intermediate inactive zone **140**.

(46) Also, as another example of the first connecting member **200**, as shown in FIG. **10**, upper and lower inactive zones, which are not filled with gas, may be formed at the ends of the lower and upper regions **110** and **120**, respectively, to serve as the first connecting member **200**.

(47) That is, the lower inactive zone **220a** is formed in a shape extending beneath the airbag cushion **100**, and the end portion of the lower inactive zone **220a** is sewn to the intermediate inactive zone **140**, while the upper inactive zone **220b** is formed in a shape extending above the airbag cushion **100**, and the end portion of the upper inactive zone **220b** is sewn to the intermediate inactive zone **140**.

(48) Meanwhile, in the present invention, the lower region **110** of the airbag cushion **100** may deploy prior to the deployment of the upper region **120**, as shown in FIG. **2**.

(49) To this end, the inflator **400** is fixed to the seat back frame of the driver's seat and connected to the lower region **110** of the airbag cushion **100**.

(50) As a result, the lower region **110** of the airbag cushion **100** deploys first to restrain the pelvis area of the occupant, which typically moves faster, and then the upper region **120** of the airbag cushion **100** deploys to restrains the head area of the occupant.

(51) Meanwhile, in the present invention, the airbag cushion **100** may be deployed not only between the driver's seat and the front passenger seat but also between the driver's seat and the inner side of the vehicle cabin, or between the front passenger seat and the inner side of the vehicle cabin.

(52) To this end, the present invention includes an airbag cushion **100** that deploys vertically from the side of the seat, a lower region **110** that deploys in a shape protruding towards the structure present beside (or positioned adjacent to) the seat from the lower part of the airbag cushion **100**, and an upper region **120** that deploys in a shape protruding towards the occupant seated on the seat from the upper part of the airbag cushion **100**.

(53) For example, the aforementioned structure may be a vehicle's side door or side body, and the seat may be a seat facing the inner side of the vehicle cabin.

(54) In the case of the front passenger seat, when the airbag is activated, the lower region **110** of the airbag cushion **100** deploys from the lower side of the seat in a shape protruding towards the front passenger side door, thereby being supported by the inner side of the front passenger side door.

(55) And the upper region **120** of the airbag cushion **100** deploys from the upper side of the seat in a shape protruding towards the occupant seated on the front passenger seat, restraining the upper body of the front passenger.

(56) Therefore, the lower region **110** of the airbag cushion **100** is supported by the vehicle's interior side, restraining the lower limbs of the occupant on the front passenger seat, and additionally, the upper region **120** of the airbag cushion **100** restrains the head of the occupant on the front passenger seat, providing protection to the occupant.

(57) Although the description is made with the front passenger seat as an example, the same concept may be applied to the driver's seat as well.

(58) Meanwhile, with reference to FIG. **12**, the present invention may be implemented with an airbag cushion including a first cushion **110'** deploying from the lower region and a second cushion **120'** deploying from the upper region, the first cushion **110'** deploying between the side of the first seat **10** and the side of the second seat **20**, the second cushion **120'** being connected to the first cushion **110'** and deploying upward from a position supporting the head of an occupant seated on

the first seat **10**, the second cushion **120'** being connected to a second connecting member **300** to create a tension to pull the second cushion **120'** towards the occupant seated on the first seat **10**.

(59) For example, the first cushion **110'** is fixed to the side of the seat cushion frame of the first seat (driver's seat) **10** facing the second seat (front passenger seat) **20**, and deploys from the side of the seat cushion **11** of the first seat **10**.

(60) As a result, the first cushion **110'** deploys towards the second seat **20** and may be supported by the side of the second seat **20**, for example, the side of the seat cushion **21** of the front passenger seat.

(61) The second cushion **120'** is connected to the upper portion of the first cushion **110'** and deploys upwards to be positioned to support the head and torso of an occupant seated on the first seat **10**.

(62) In addition, the second connecting member **300** is connected to create a tension to pull the second cushion **120'** towards the occupant on the driver's seat when the airbag is activated.

(63) Accordingly, in a vehicle without a center console, when a passenger is loaded onto the second cushion **120'**, the first cushion **110'** being supported by the second seat **20** prevents the second cushion **120'** from rotating and resultantly preventing the upper body of the occupant on the driver's seat from shifting towards the front passenger side, leading to enhancement of driver's restraint.

(64) Furthermore, as the second cushion **120'** is pulled towards the occupant on the driver's seat by the second connecting member **300**, it more effectively restrains the head of the driver, providing enhanced protection to the occupant.

(65) Additionally, in the present invention, the first cushion **110'** may be supported by the seat cushion **11** of the first seat **10**.

(66) That is, in vehicles without a center console, there is a problem where it becomes difficult to effectively restrain the lower limbs of the occupant using the airbag cushion due to the lack of proper support for the airbag cushion.

(67) In this regard, in the present invention, the first cushion **110'** is fixed to the seat cushion **11** of the driver's seat, and as a result, the first cushion **110'** is supported by the seat cushion **21** of the front passenger seat, allowing the lower limbs of the driver to be supported by the first cushion **110'** deployed from the seat cushion **11** of the driver's seat.

(68) Particularly, the first cushion **110'** may be configured to deploy from the rear side of the seat cushion **11**.

(69) With reference to FIG. **13**, the first cushion **110'** may be deployed from the rear side of the seat cushion **11** towards the front for a predetermined length.

(70) That is, when an occupant is seated on the first seat **10**, the pelvis of the occupant on the driver's seat, being in contact with the lower part of the seat back, is positioned at the rear end of the seat cushion **11** where it meets the seat back.

(71) Therefore, with the deployment of the first cushion **110'** from the rear end of the seat cushion **11**, the first cushion **110'** enhances the pelvic support for the occupant on the driver's seat.

(72) Meanwhile, in the present invention, the second cushion **120'** may be connected to the position of the first cushion **110'** that is biased towards the first seat **10**.

(73) For example, the first cushion **110'** is made of one panel **100a** facing the driver's side and another panel **100b** facing the front passenger, which are sewn at their edges together.

(74) That is, the lower portion of the second cushion **120'** is connected to the upper part of the panel **100a** that deploys towards the first seat **10** in a sewn structure.

(75) Therefore, the second cushion **120'** deploys from a position relatively closer to the occupant on the driver's seat rather than the front passenger seat, allowing for a quicker restraint of the driver.

(76) Additionally, the first cushion **110'** may be configured to deploy prior to the second cushion **120'**.

(77) To achieve this, an inflater **400** is connected to the first cushion **110'**, and the inflater **400** is mounted and secured to the seat cushion frame of the first seat **10**.

(78) As a result, the first cushion **110'** deploys prior to the second cushion **120'**, restraining the pelvic region of the occupant that experiences relatively faster movement, and subsequently, the second cushion (**120'**) deploys, restraining the head region of the occupant.

(79) In addition, the second cushion **120'** may be designed to have a smaller deployment capacity compared to the first cushion **110'**.

(80) That is, the first cushion **110'** serves as a substitute for the center console, so the capacity of the first cushion **110'** has to be relatively larger compared to the second cushion **120'**.

(81) Meanwhile, the second cushion **120'** for protecting the occupant's head is designed to have a relatively smaller capacity compared to the first cushion **110'**, allowing the second cushion **120'** to deploy more rapidly.

(82) By reducing the deployment time of the second cushion **120'**, the protective area provided by the second cushion **120'** may be established quickly, allowing for the rapid and secure restraint and protection of the occupant.

(83) In addition, the first cushion **110'** and the second cushion **120'** are interconnected, allowing for the flow of gas therebetween.

(84) That is, the inflator **400** is connected to the first cushion **110'** to supply gas to the interior of the first cushion **110'**, and the first cushion **110'** is interconnected with the second cushion **120'**, allowing gas to flow into the interior of the second cushion **120'**.

(85) Here, the second cushion **120'** should be formed to be larger than the passage connecting the first cushion **110'** and the second cushion **120'** in order to cover the passage. In order to reduce the deployment volume of the second cushion **120'**, it is preferable to shape the cross-sectional area of the passage as narrow as possible.

(86) Meanwhile, the second connecting member **300** may be connected to the position where the occupant of the first seat **10** is supported when the airbag is deployed.

(87) That is, when the occupant on the driver's seat moves towards the front passenger side due to a collision and comes into contact with the second connecting member **300**, a tension is generated in the second connecting member **300** as the load from the occupant on the driver's seat is applied to the second connecting member **300**.

(88) As a result, the second cushion **120'** moves towards the occupant on the driver's seat, allowing for early restraint of the head of the occupant.

(89) Furthermore, the second connecting member **300** may be connected between the first cushion **110'** and the second cushion **120'**.

(90) Preferably, the second connecting member **300** may be connected between the side of the first cushion **110'** facing the occupant on the driver's seat **10** and the top of the second cushion **120'**.

(91) In detail, the top end of the second connecting member **300** is sewn to the topmost part of the second cushion **120'**, while the bottom end of the second connecting member **300** is sewn to the middle of the panel facing the driver's seat among the two panels of the first cushion **110'**.

(92) Therefore, as the second connecting member **300** spreads out towards the occupant on the driver's seat, the occupant on the driver's seat may make direct contact with the second connecting member **300** securely.

(93) As an example of the second connecting member **300**, a tether may be employed.

(94) The tether **300** is formed in the shape of a belt or band of which the top end is sewn to the second cushion **120'** and the bottom end is sewn to the first cushion **110'**.

(95) Meanwhile, the airbag cushion of the present invention may be applied as a center side airbag that deploys between the driver's seat and the front passenger seat and be configured to deploy between the driver's seat and the vehicle's interior side or between the front passenger seat and the vehicle's interior side.

(96) With reference to FIG. **14**, the present invention comprises a first cushion **110'** deploying and being supported between a side of a seat and a structure beside the seat, a second cushion **120'** connected to the first cushion **110'** and deploying upward from a position supporting the head of the

occupant seated on the seat, and a second connecting member **300** connected to the second cushion **120'** to create a tension to pull the second cushion **120'** towards the occupant seated on the seat.

(97) For example, the aforementioned structure may be a vehicle's side door or side body, and the seat may be a seat facing the inner side of the vehicle cabin.

(98) In the case of the second seat **20**, when the airbag is activated, the first cushion **110'** deploys from the side of the seat cushion **21** and takes on a shape protruding towards the front passenger side door to be supported by the inner side of the front passenger side door.

(99) The second cushion **120'** deploys from the upper position of the first cushion **110'** to support the head of the occupant on the front passenger seat.

(100) Furthermore, the second connecting member **300** is connected in a manner that creates a tension to pull the second cushion **120'** towards the occupant on the front passenger seat when the airbag is activated.

(101) Therefore, the first cushion **110'** is supported against the inner side of the vehicle cabin, restraining the legs of the occupant on the front seat, and simultaneously, the second cushion **120'** restrains the head of the occupant on the front passenger seat, providing protection.

(102) Although the description is made with the second seat as an example, the same concept may be applied to the first seat **10** as well.

(103) Hereinafter, a description is made of the method for assembling the airbag cushion **100** of the center side airbag as depicted in FIGS. **9** and **10**.

(104) The lower inactive zone **220a** or tether **210** formed at the lower end of the airbag cushion **100** is folded outward from the airbag cushion **100**, and the end of the lower inactive zone **220a** or the tether **210** is sewn and fixed to the intermediate inactive zone **140** located at the center portion on the outer surface of the airbag cushion **100**.

(105) Next, the upper inactive zone **220b** or tether **210** formed at the upper end of the airbag cushion **100** is folded inward from the airbag cushion **100**, and the end of the upper inactive zone **220b** or the tether **210** is sewn and fixed to the intermediate inactive zone **140** located at the center portion on the inner surface of the airbag cushion **100**.

(106) As a result, when the airbag is activated, the lower portion of the airbag cushion **100** deploys in a shape protruding towards the front passenger seat, being supported by the side of the front passenger seat.

(107) Furthermore, the upper portion of the airbag cushion **100** deploys in a shape protruding towards the occupant seated on the driver's seat, providing support to the upper body of the driver.

(108) Therefore, in vehicles without a center console, the lower region **110** of the airbag cushion **100** is supported on the front passenger seat, restraining the lower body of the driver, while the upper region **120** of the airbag cushion **100** restrains the head of the driver, providing protection to the driver.

(109) In addition, as the airbag cushion **100** is supported by the front passenger seat, it prevents the airbag cushion **100** from rotating, thus preventing the upper body of the driver from moving towards the front passenger side, enhancing occupant restraint.

(110) Furthermore, by providing support to the occupants through the shape modification of the airbag cushion alone, there is no need to increase the volume or capacity of the inflator, resulting in cost and weight reduction of the airbag cushion and offering advantages in terms of airbag packaging.

(111) Although the present invention has been described in detail with specific examples, but it is obvious to those skilled in the art that various modifications and alteration can be made without departing from the scope of the technical idea of this invention, and such modifications and alterations fall within the scope of the appended claims.

Claims

1. An airbag configured, when deployed, to be positioned between first and second seats disposed side by side in a vehicle, the airbag having an airbag cushion comprising: an intermediate region configured to expand between the first and second seats when the airbag cushion is inflated and deployed; a lower region connected to a lower end portion of the intermediate region and configured to protrude from the lower end portion of the intermediate region toward the second seat when the airbag cushion is inflated and deployed; and an upper region connected to an upper end portion of the intermediate region and configured, when deployed, to protrude from the upper end portion of the intermediate region toward an occupant seated on the first seat when the airbag cushion is inflated and deployed, wherein, when the airbag cushion is inflated and deployed between the first and second seats, the intermediate region has a first length in a vertical direction of the vehicle that is greater than a second length of the lower region in the vertical direction of the vehicle and a third length of the upper region in the vertical direction of the vehicle.
 2. The airbag of claim 1, wherein the lower region is configured to be bent to protrude toward the second seat when the airbag cushion is inflated and deployed.
 3. The airbag of claim 2, further comprising a connecting member connected to and extending between the lower region and the intermediate region and configured to create a tension causing the lower region to be bent.
 4. The airbag of claim 3, wherein: the intermediate region includes an inactive zone configured not to be filled with gas when the airbag cushion is inflated, and the connecting member is connected to the inactive zone.
 5. The airbag of claim 3, wherein the connecting member comprises upper and lower inactive zones respectively extending from the upper and lower regions.
 6. The airbag of claim 1, wherein the upper region is configured to be bent to protrude toward the first seat when the airbag cushion is inflated and deployed.
 7. The airbag of claim 6, further comprising a connecting member extending between the upper region and the intermediate region and configured to create a tension causing the upper region to be bent.
 8. The airbag of claim 1, wherein the lower region is configured to deploy before the upper region deploys when the airbag cushion is inflated and deployed.
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